

Robert Harrison

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PROFESSIONAL SUMMARY

Senior Software Engineer with 9+ years of experience designing and building scalable, full-stack, and distributed systems, most recently at Tesla. Expertise across React/TypeScript frontends, C#.NET backends, REST APIs, microservices, event-driven architecture, PostgreSQL, Kafka, RabbitMQ, Docker, and Kubernetes. Experienced in operating mission-critical applications with a focus on reliability, performance, and maintainable system design.

PROFESSIONAL EXPERIENCE

Tesla, Inc. | Oct 2020 – Apr 2024

Senior Software Engineer

- Primary owner of Tesla's Energy Pricing Gateway for both Residential and Commercial, handling all pricing requests for Solar Roof, Solar Panels, Powerwall, and Megapack.
- Built and supported Camunda-orchestrated workflows for Tesla Energy residential orders, enabling operations teams to look up an order ID and see the active workflow token showing exactly where each solar job was in the pipeline.
- Engineered the System Cost Update Tool to automatically reprice 5,000+ active quotes across a multi-billion-dollar Megapack pipeline, eliminating ~90% of manual pricing effort via a time-boxed, batched processing workflow built to handle CPU- and memory-intensive calculations within service timeout limits.
- Automated Megapack delivery scheduling with a full-stack application (React, .NET, PostgreSQL, Kafka) that ingests upstream factory build schedules and notifies downstream logistics teams and project contacts – delivering major time savings by eliminating a fully manual process.
- Improved production observability by implementing service-health dashboards, monitoring, and automated alerting, enabling faster detection and diagnosis of production issues.
- Led design discussions with Product whenever new capabilities were scoped for the commercial quoting platform, translating requirements into detailed technical plans of action, and regularly mentored junior engineers on service architecture and implementation approach.

LBA Ware (Acquired by SimpleNexus, an nCino Company) | Jan 2019 – Oct 2020

Software Developer

- Developed high-performance BI dashboards using Angular, TypeScript, C#.NET, Redis, and SQL Server, enabling mortgage companies to analyze operational and performance metrics while LBA Ware scaled rapidly and was ultimately acquired.
- Implemented Microsoft OAuth-based SSO, enabling users to securely authenticate with their existing Microsoft credentials and streamlining access across the platform.

Citrix | Jul 2013 – Nov 2016

Software Engineer

- Delivered the Azure Provisioning Plugin for XenDesktop (WPF/XAML), using Azure APIs to automatically migrate and scale customers' on-premises deployments into Microsoft Azure – provisioning and managing the underlying virtual desktop infrastructure end-to-end with no manual setup required.

TECHNICAL SKILLS

Languages: JavaScript, C#, TypeScript, SQL

Frontend: React

Backend, Architecture, & Infrastructure: Node.js, .NET / ASP.NET Core, REST APIs, Websockets, RPC, SSE, Microservices, Distributed Systems, Event-Driven Architecture, Camunda, Docker, Kubernetes

Data & Messaging: PostgreSQL, Redis, Dapper, Entity Framework Core, Kafka, RabbitMQ, Stored Procedures

EDUCATION

Georgia Institute of Technology | 2010 – 2013

Bachelor of Science, Computer Science | Highest Honors | GPA: 3.6/4.0

PROFESSIONAL DEVELOPMENT

Formation — Engineering Fellow

Aug 2026 – Present

- Advanced practice in system design, algorithms, and technical communication with mentorship from staff-level engineers.